

Experiment 07:

7. Laboratory Exercise

index.html code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Node.js Server</title>
</head>
<body>
  <h1>Welcome to My Node.js Server!</h1>
  <p>This file was handled and served using the built-in fs module.</p>
</body>
</html>
```

Server.js code:

```
// Step 1: Import the core Node.js modules
const http = require('http');
const fs = require('fs');
const path = require('path');

const PORT = 3000;

// Step 2: Create the HTTP server
const server = http.createServer((req, res) => {

  // Log the request to a local file (Asynchronous File Appending)
  const logMessage = `[${new Date().toISOString()}] Request received for:
${req.url}\n`;
  fs.appendFile('server.log', logMessage, (err) => {
    if (err) console.error('Failed to write to log file:', err);
  });

  // Step 3: Handle routes and serve the HTML file
  if (req.url === '/' || req.url === '/index.html') {

    // Read the index.html file asynchronously
    fs.readFile(path.join(__dirname, 'index.html'), 'utf8', (err, data) =>
    {
      if (err) {
        // Handle file system error (e.g., file missing)
        res.writeHead(500, { 'Content-Type': 'text/plain' });
        res.end('500 Internal Server Error: Missing HTML source.');
```

```
// Send successful response with HTML content
res.writeHead(200, { 'Content-Type': 'text/html' }); // Tells
browser it is HTML
res.end(data);
});

} else {
  // Fallback for any other page routes
  res.writeHead(404, { 'Content-Type': 'text/plain' });
  res.end('404 Page Not Found');
}
});

// Step 4: Start listening on the designated port
server.listen(PORT, () => {
  console.log(`Server is actively running at http://localhost:${PORT}/`);
});
```

Output:

